

PHASE 1 — CORE NUMERICAL THINKING

Arithmetic & Quantitative Fluency

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Mental arithmetic
 - ☐ Fast addition/subtraction
 - ☐ Multiplication patterns
 - ☐ Division estimation
 - ☐ Percentage calculations
 - ☐ Ratio intuition
 - ☐ Approximation techniques
 - ☐ Order-of-magnitude estimation
 - ☐ Unit scaling intuition
- ☐ Fractions, decimals, percentages
 - ☐ Fraction simplification
 - ☐ Decimal conversion
 - ☐ Percentage change
 - ☐ Compound percentages
 - ☐ Financial calculations
- ☐ Scientific notation
 - ☐ Very large numbers
 - ☐ Very small numbers
 - ☐ Engineering notation
 - ☐ Power-of-10 estimation
- ☐ Dimensional thinking
 - ☐ Units conversion
 - ☐ SI units
 - ☐ Dimensional consistency

- ☐ Unit sanity checking
- ☐ Practical engineering calculations
 - ☐ Voltage/current/power
 - ☐ RPM calculations
 - ☐ Torque estimation
 - ☐ Speed-distance-time
 - ☐ Efficiency calculations
 - ☐ Battery runtime estimation
 - ☐ Memory/storage calculations
- ☐ Financial mathematics basics
 - ☐ Revenue calculations
 - ☐ Margin calculations
 - ☐ Break-even analysis
 - ☐ Compound growth
 - ☐ CAGR intuition
 - ☐ Pricing math
 - ☐ ROI calculations

Mathematical Communication

(priority: High)

(depth: Working Knowledge)

- ☐ Reading equations comfortably
- ☐ Translating real systems into variables
- ☐ Rearranging formulas
- ☐ Variable isolation
- ☐ Interpreting graphs
- ☐ Engineering notation usage
- ☐ Spreadsheet-based modeling

Practical Tools

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Spreadsheet mathematics
 - ☐ Microsoft Excel
 - ☐ Google Sheets
 - ☐ Formula writing

- ☐ Lookup functions
- ☐ Conditional logic
- ☐ Pivot tables
- ☐ Data cleaning
- ☐ Scenario modeling
- ☐ Calculator tools
 - ☐ Scientific calculator usage
 - ☐ Graphing calculators
 - ☐ Engineering calculator apps
- ☐ Computational tools
 - ☐ Python math libraries
 - ☐ NumPy basics
 - ☐ SymPy basics
 - ☐ Jupyter notebooks
 - ☐ Desmos graphing
 - ☐ WolframAlpha usage

PHASE 2 — ALGEBRA FOR BUILDERS

Algebra Fundamentals

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Variables and equations
 - ☐ Linear equations
 - ☐ Multi-variable equations
 - ☐ Formula manipulation
 - ☐ Engineering equation solving
- ☐ Functions
 - ☐ Input-output thinking
 - ☐ Function graphs
 - ☐ Piecewise functions
 - ☐ Domain/range intuition
- ☐ Linear relationships
 - ☐ Slope interpretation
 - ☐ Rate-of-change thinking

- ☐ Calibration equations
- ☐ Sensor scaling
- ☐ Systems of equations
 - ☐ Two-variable systems
 - ☐ Matrix-based solving
 - ☐ Constraint modeling
 - ☐ Optimization intuition
- ☐ Exponents & logarithms
 - ☐ Growth modeling
 - ☐ Exponential decay
 - ☐ Battery discharge models
 - ☐ Signal attenuation
 - ☐ Log-scale interpretation
 - ☐ pH/dB/logarithmic systems
- ☐ Polynomial intuition
 - ☐ Quadratics
 - ☐ Curve fitting basics
 - ☐ Physical system approximation

Applied Algebra

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Translating real systems into equations
- ☐ Engineering constraints
- ☐ Budget modeling
- ☐ Cost optimization
- ☐ Sensor conversion formulas
- ☐ Calibration equations
- ☐ Data trend equations
- ☐ Performance estimation
- ☐ Throughput calculations
- ☐ Resource allocation math

Mathematical Modeling

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Simplifying complex systems
- ☐ Assumption-based modeling
- ☐ Approximation methods
- ☐ Error estimation
- ☐ Tradeoff analysis
- ☐ Scaling laws
- ☐ Sensitivity analysis

Project Checkpoints

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Build a cost calculator
- ☐ Build a battery runtime estimator
- ☐ Build a robotics gear-ratio calculator
- ☐ Build a startup financial model
- ☐ Build a sensor calibration spreadsheet
- ☐ Build a productivity analytics dashboard

PHASE 3 — GEOMETRY & SPATIAL THINKING

Practical Geometry

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Coordinate geometry
 - ☐ Cartesian systems
 - ☐ Distance calculations
 - ☐ Midpoint calculations
 - ☐ Angle interpretation
- ☐ Measurement geometry
 - ☐ Area calculations
 - ☐ Volume calculations
 - ☐ Surface area
 - ☐ Packing efficiency
- ☐ Trigonometry fundamentals
 - ☐ Sine/cosine/tangent

- ☐ Angle calculations
- ☐ Vector decomposition
- ☐ Slope interpretation
- ☐ Navigation calculations
- ☐ Spatial visualization
 - ☐ 2D to 3D reasoning
 - ☐ Rotations
 - ☐ Transformations
 - ☐ Coordinate transformations
- ☐ Engineering geometry
 - ☐ CAD dimensioning
 - ☐ Mechanical tolerances
 - ☐ Robot arm positioning
 - ☐ Linkage geometry

CAD & Geometry Integration

(priority: High)

(depth: Working Knowledge)

- ☐ Fusion 360 basics
- ☐ Sketch constraints
- ☐ Parametric modeling
- ☐ Measurement precision
- ☐ 3D printing tolerances
- ☐ CNC coordinate understanding

Robotics Geometry

(priority: High)

(depth: Working Knowledge)

- ☐ Forward kinematics intuition
- ☐ Coordinate frames
- ☐ Motion geometry
- ☐ Wheel odometry basics
- ☐ Camera geometry basics

Project Checkpoints

(priority: High)

(depth: Working Knowledge)

- ☐ Design a parametric CAD model
- ☐ Build a trigonometric robot-angle calculator
- ☐ Create a room-mapping system
- ☐ Simulate robotic arm movement

PHASE 4 — APPLIED TRIGONOMETRY & VECTORS

Trigonometry for Engineering

(priority: High)

(depth: Deep Mastery)

- ☐ Right-triangle trigonometry
- ☐ Unit circle intuition
- ☐ Periodic systems
- ☐ Oscillation interpretation
- ☐ Wave intuition
- ☐ Rotational systems
- ☐ Phase understanding

Vector Mathematics

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Vector basics
 - ☐ Magnitude
 - ☐ Direction
 - ☐ Components
- ☐ Vector operations
 - ☐ Addition/subtraction
 - ☐ Dot product
 - ☐ Cross product intuition
- ☐ Applied vectors
 - ☐ Forces
 - ☐ Velocity
 - ☐ Acceleration
 - ☐ Motion planning

- ☐ Navigation systems
- ☐ Coordinate systems
 - ☐ Local/global coordinates
 - ☐ Rotational transformations
 - ☐ Reference frames

Applied Physics Math

(priority: High)

(depth: Working Knowledge)

- ☐ Motion equations
- ☐ Rotational motion
- ☐ Mechanical force calculations
- ☐ Electrical waveforms
- ☐ Signal interpretation

PHASE 5 — CALCULUS FOR REAL SYSTEMS

Calculus Intuition

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Rate-of-change intuition
- ☐ Accumulation intuition
- ☐ Continuous systems thinking
- ☐ Change over time

Differential Calculus

(priority: Extremely High)

(depth: Working Knowledge)

- ☐ Derivative intuition
- ☐ Slopes & rates
- ☐ Optimization problems
- ☐ Max/min analysis
- ☐ Velocity/acceleration
- ☐ Sensitivity analysis
- ☐ Practical applications

- ☐ Robotics motion
- ☐ Motor speed changes
- ☐ Data trend analysis
- ☐ System response understanding

Integral Calculus

(priority: High)

(depth: Working Knowledge)

- ☐ Area-under-curve intuition
- ☐ Accumulation systems
- ☐ Energy estimation
- ☐ Signal accumulation
- ☐ Resource consumption estimation

Differential Equations

(priority: High)

(depth: Working Knowledge)

- ☐ First-order intuition
- ☐ Feedback system behavior
- ☐ Exponential response systems
- ☐ PID intuition
- ☐ Dynamic system understanding

Practical Computational Calculus

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Numerical methods
 - ☐ Numerical differentiation
 - ☐ Numerical integration
 - ☐ Simulation loops
- ☐ Simulation tools
 - ☐ Python scientific stack
 - ☐ MATLAB awareness
 - ☐ Simulink awareness
 - ☐ SciPy basics

Project Checkpoints

(priority: High)

(depth: Working Knowledge)

- ☐ Simulate motor acceleration
- ☐ Build a PID tuning visualizer
- ☐ Build a motion prediction model
- ☐ Create a simple physics engine

PHASE 6 — LINEAR ALGEBRA FOR MODERN SYSTEMS

Core Linear Algebra

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Matrices
 - ☐ Matrix operations
 - ☐ Matrix multiplication
 - ☐ Inverses intuition
- ☐ Vectors in higher dimensions
- ☐ Systems of equations
- ☐ Transformations
- ☐ Coordinate transformations

Practical Applications

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Robotics transformations
- ☐ 3D graphics math
- ☐ Computer vision basics
- ☐ ML tensor intuition
- ☐ State-space systems
- ☐ Signal processing basics

Numerical Computing

(priority: High)

(depth: Working Knowledge)

- ☐ NumPy matrix operations
- ☐ Efficient array operations
- ☐ Tensor basics
- ☐ GPU computation awareness

AI/ML Math Foundations

(priority: High)

(depth: Working Knowledge)

- ☐ Vector spaces intuition
- ☐ Embeddings intuition
- ☐ Gradient-based optimization
- ☐ Linear regression math
- ☐ Dimensionality reduction intuition

Project Checkpoints

(priority: High)

(depth: Working Knowledge)

- ☐ Build a 3D transformation simulator
- ☐ Build a linear regression model
- ☐ Simulate robot coordinate transforms
- ☐ Create matrix-based animations

PHASE 7 — PROBABILITY, STATISTICS & DATA THINKING

Probability Fundamentals

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Probability intuition
- ☐ Conditional probability
- ☐ Independent events
- ☐ Risk estimation
- ☐ Failure probability

Statistics for Builders

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Mean/median/mode
- ☐ Variance & standard deviation
- ☐ Distributions intuition
- ☐ Sampling intuition
- ☐ Noise understanding
- ☐ Data analysis
 - ☐ Outlier detection
 - ☐ Trend analysis
 - ☐ Correlation
 - ☐ Regression basics

Experimental Thinking

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ A/B testing basics
- ☐ Product metrics
- ☐ KPI interpretation
- ☐ Statistical decision-making
- ☐ Confidence intuition

Sensor & Signal Statistics

(priority: High)

(depth: Working Knowledge)

- ☐ Sensor noise
- ☐ Filtering basics
- ☐ Averaging techniques
- ☐ Moving averages
- ☐ Kalman filter awareness

Data Tools

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Pandas basics
- ☐ Data visualization

- ☐ CSV data workflows
- ☐ Dashboarding basics
- ☐ Time-series interpretation

Project Checkpoints

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Analyze sensor datasets
- ☐ Build startup KPI dashboards
- ☐ Create anomaly detectors
- ☐ Build forecasting spreadsheets
- ☐ Build product analytics systems

PHASE 8 — DISCRETE MATH FOR PROGRAMMERS & SYSTEM DESIGN

Logic & Boolean Thinking

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Boolean algebra
- ☐ Logical operators
- ☐ Truth tables
- ☐ Conditional logic
- ☐ Binary reasoning

Set Theory Basics

(priority: Medium)

(depth: Working Knowledge)

- ☐ Sets
- ☐ Intersections
- ☐ Unions
- ☐ Membership logic

Graphs & Networks

(priority: High)

(depth: Working Knowledge)

- ☐ Graph intuition
- ☐ Nodes/edges
- ☐ Network routing basics
- ☐ Dependency systems
- ☐ Relationship modeling

Combinatorics Basics

(priority: Medium)

(depth: Working Knowledge)

- ☐ Counting methods
- ☐ Permutations
- ☐ Combinations
- ☐ Search-space intuition

Binary & Digital Math

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Binary numbers
- ☐ Hexadecimal
- ☐ Bitwise operations
- ☐ Data representation
- ☐ Encoding basics

Algorithmic Thinking

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Complexity intuition
- ☐ Scaling analysis
- ☐ Optimization tradeoffs
- ☐ Recursive thinking
- ☐ State-machine thinking

PHASE 9 — OPTIMIZATION & SYSTEMS THINKING

Optimization

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Constraint optimization
- ☐ Cost minimization
- ☐ Resource allocation
- ☐ Scheduling logic
- ☐ Throughput optimization
- ☐ Bottleneck analysis

Operations & Startup Mathematics

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Unit economics
- ☐ Customer acquisition math
- ☐ Inventory mathematics
- ☐ Supply chain calculations
- ☐ Capacity planning
- ☐ Queue intuition

Engineering Tradeoff Mathematics

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Cost vs performance
- ☐ Speed vs accuracy
- ☐ Precision vs computation
- ☐ Power vs efficiency
- ☐ Reliability vs complexity

Simulation & Modeling

(priority: High)

(depth: Working Knowledge)

- ☐ Monte Carlo awareness
- ☐ System simulation

- ☐ Scenario analysis
- ☐ Forecasting systems

PHASE 10 — APPLIED COMPUTATIONAL MATHEMATICS

Programming-Oriented Mathematics

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Numerical programming
- ☐ Floating-point awareness
- ☐ Precision errors
- ☐ Overflow/underflow
- ☐ Efficient computation

Scientific Computing Stack

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ NumPy
- ☐ SciPy
- ☐ Matplotlib
- ☐ Pandas
- ☐ Jupyter workflows

Visualization Mathematics

(priority: High)

(depth: Working Knowledge)

- ☐ Data visualization
- ☐ Plot interpretation
- ☐ Signal visualization
- ☐ Engineering dashboard creation

Simulation Tools

(priority: High)

(depth: Working Knowledge)

- ☐ MATLAB awareness
- ☐ GNU Octave
- ☐ Simulink awareness
- ☐ Physics simulators

Robotics & Automation Math Integration

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Sensor fusion intuition
- ☐ Motion estimation
- ☐ Path planning basics
- ☐ Control-system math
- ☐ Embedded-system calculations

PHASE 11 — HIGH-ROI SPECIALIZED TOPICS

Signal Processing Basics

(priority: High)

(depth: Working Knowledge)

- ☐ Frequency intuition
- ☐ Fourier transform awareness
- ☐ Filtering basics
- ☐ Noise reduction intuition

Control Systems Math

(priority: High)

(depth: Working Knowledge)

- ☐ Feedback loops
- ☐ PID mathematics
- ☐ Stability intuition
- ☐ System response graphs

Machine Learning Mathematics

(priority: Medium)

(depth: Working Knowledge)

- ☐ Gradient descent intuition
- ☐ Loss functions
- ☐ Optimization basics
- ☐ Neural network matrix intuition

Cryptography Awareness

(priority: Optional)

(depth: Awareness Only)

- ☐ Modular arithmetic basics
- ☐ Hash function intuition
- ☐ Encryption awareness

Computer Graphics Mathematics

(priority: Medium)

(depth: Working Knowledge)

- ☐ 3D transformations
- ☐ Projection mathematics
- ☐ Camera mathematics
- ☐ Rendering intuition

PHASE 12 — DEBUGGING, VALIDATION & ENGINEERING INTUITION

Mathematical Debugging

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Sanity checking
- ☐ Unit verification
- ☐ Range estimation
- ☐ Error bounding
- ☐ Edge-case analysis

Estimation Skills

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Fermi estimation
- ☐ Quick engineering approximations
- ☐ Back-of-envelope calculations
- ☐ Feasibility estimation

Real-World Validation

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Compare theory vs measurement
- ☐ Experimental calibration
- ☐ Measurement uncertainty
- ☐ Data-driven iteration

Systems Thinking

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Multi-variable interaction
- ☐ Feedback systems
- ☐ Scaling effects
- ☐ Failure-mode reasoning
- ☐ Constraint-aware thinking

OPTIONAL / LOW-ROI TOPICS

Low Practical ROI Academic Topics

(priority: Optional)

(depth: Awareness Only)

- ☐ Formal proof techniques
- ☐ Abstract algebra
- ☐ Real analysis
- ☐ Complex analysis
- ☐ Topology
- ☐ Number theory

- ☐ Measure theory
- ☐ Tensor calculus
- ☐ Advanced PDE theory
- ☐ Pure combinatorics
- ☐ Abstract geometry
- ☐ Category theory
- ☐ Advanced symbolic integration
- ☐ Advanced theorem-heavy calculus
- ☐ Academic epsilon-delta proofs



Create a FULLY DETAILED practical-first CHECKLIST
roadmap for: MECHANICAL FOR MECHATRONICS

IMPORTANT:

This roadmap is NOT for academic specialization or degree completion.

It is for a:

- multidisciplinary builder,
- startup founder,
- technical entrepreneur,
- hobbyist engineer,
- robotics/mechatronics learner,
- automation-focused creator,
- practical systems builder.

The goal is NOT:

- complete mastery of every branch,
- academic perfection,
- research-level depth,
- exam preparation.

The goal IS:

- maximum practical capability,
- highest real-world ROI,

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PHASE 1 — CORE MECHANICAL FOUNDATIONS

Engineering Intuition & Physical Thinking

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Force intuition
 - ☐ Push/pull mechanics
 - ☐ Load paths
 - ☐ Mechanical advantage
 - ☐ Constraint understanding
- ☐ Motion intuition
 - ☐ Linear motion
 - ☐ Rotational motion
 - ☐ Speed vs torque
 - ☐ Inertia intuition
 - ☐ Momentum intuition
- ☐ Energy transfer intuition
 - ☐ Mechanical efficiency
 - ☐ Friction losses
 - ☐ Heat generation
 - ☐ Vibration sources
- ☐ Structural intuition
 - ☐ Rigidity vs flexibility
 - ☐ Stress concentration awareness
 - ☐ Failure intuition
 - ☐ Stability thinking
- ☐ Mechanical system decomposition
 - ☐ Input-output thinking

- ☐ Transmission systems
- ☐ Support structures
- ☐ Motion conversion systems

Practical Mathematics for Mechanical Systems

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Unit conversions
- ☐ Force calculations
- ☐ Torque calculations
- ☐ Gear ratio calculations
- ☐ RPM calculations
- ☐ Speed-distance-time calculations
- ☐ Power calculations
- ☐ Efficiency calculations
- ☐ Trigonometry for mechanisms
- ☐ Basic vector intuition

Mechanical Drawing & Visualization

(priority: High)

(depth: Working Knowledge)

- ☐ Orthographic views
- ☐ Isometric visualization
- ☐ Dimension interpretation
- ☐ Tolerance reading
- ☐ Section views
- ☐ Exploded assembly understanding

PHASE 2 — MECHANICAL HARDWARE BASICS

Fasteners & Assembly Systems

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Screws & bolts
 - ☐ Metric fasteners

- ☐ Thread types
- ☐ Bolt sizing
- ☐ Thread pitch
- ☐ Nuts & washers
 - ☐ Locking mechanisms
 - ☐ Load distribution
 - ☐ Anti-vibration fastening
- ☐ Threaded inserts
 - ☐ Heat-set inserts
 - ☐ Rivnuts
 - ☐ Tapped holes
- ☐ Assembly techniques
 - ☐ Torque tightening
 - ☐ Thread lockers
 - ☐ Alignment methods
 - ☐ Repeatable assembly
- ☐ Structural joining
 - ☐ Rivets
 - ☐ Adhesives
 - ☐ Welding awareness
 - ☐ Mechanical brackets

Bearings & Motion Support

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Bearing fundamentals
 - ☐ Radial bearings
 - ☐ Thrust bearings
 - ☐ Linear bearings
- ☐ Bearing selection
 - ☐ Load ratings
 - ☐ RPM considerations
 - ☐ Friction considerations
 - ☐ Mounting methods
- ☐ Shaft support systems
 - ☐ Pillow blocks

- ☐ Couplers
- ☐ Alignment methods
- ☐ Linear motion systems
 - ☐ Linear rails
 - ☐ Guide rods
 - ☐ Bushings
 - ☐ Drawer-slide mechanisms

Springs & Elastic Components

(priority: High)

(depth: Working Knowledge)

- ☐ Compression springs
- ☐ Tension springs
- ☐ Torsion springs
- ☐ Dampers
- ☐ Shock absorption
- ☐ Compliance systems

Mechanical Power Transmission

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Gears
 - ☐ Spur gears
 - ☐ Planetary gears
 - ☐ Worm gears
 - ☐ Gear ratios
- ☐ Belt systems
 - ☐ Timing belts
 - ☐ Pulley systems
 - ☐ Belt tensioning
- ☐ Chain drives
 - ☐ Sprockets
 - ☐ Chain alignment
 - ☐ Tensioning systems
- ☐ Couplers
 - ☐ Rigid couplers

- ☐ Flexible couplers
- ☐ Lead screws & ball screws
 - ☐ Linear actuation
 - ☐ Backlash awareness
 - ☐ Precision motion systems

PHASE 3 — MATERIALS FOR BUILDERS

Engineering Materials

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Aluminum
 - ☐ 6061 basics
 - ☐ Lightweight structures
 - ☐ Machinability
- ☐ Steel
 - ☐ Mild steel
 - ☐ Stainless steel
 - ☐ Strength vs weight
- ☐ Plastics
 - ☐ PLA
 - ☐ PETG
 - ☐ ABS
 - ☐ Nylon
 - ☐ Delrin/POM
 - ☐ Polycarbonate
- ☐ Composite awareness
 - ☐ Carbon fiber basics
 - ☐ Fiberglass basics

Material Selection

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Strength vs cost
- ☐ Weight optimization

- ☐ Corrosion resistance
- ☐ Heat resistance
- ☐ Flexibility vs rigidity
- ☐ Manufacturability
- ☐ Vibration behavior

Failure Awareness

(priority: High)

(depth: Working Knowledge)

- ☐ Fatigue basics
- ☐ Cracking awareness
- ☐ Buckling intuition
- ☐ Wear mechanisms
- ☐ Thermal expansion awareness

PHASE 4 — CAD & DIGITAL MECHANICAL DESIGN

CAD Fundamentals

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Sketching
 - ☐ Constraints
 - ☐ Dimensions
 - ☐ Parametric design
- ☐ 3D modeling
 - ☐ Extrusions
 - ☐ Revolves
 - ☐ Sweeps
 - ☐ Lofting
- ☐ Assemblies
 - ☐ Joints
 - ☐ Motion constraints
 - ☐ Interference checking
- ☐ Drawings
 - ☐ Manufacturing drawings

- ☐ Tolerancing
- ☐ Assembly drawings

CAD Platforms

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Fusion 360
- ☐ SolidWorks awareness
- ☐ FreeCAD awareness
- ☐ CAD file management
- ☐ Version control for CAD

Design for Manufacturing (DFM)

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Manufacturable geometry
- ☐ Simplified assembly
- ☐ Part count reduction
- ☐ Tolerance-aware design
- ☐ Tool accessibility
- ☐ Fastener accessibility
- ☐ Modular design thinking

Mechanical Simulation Awareness

(priority: Medium)

(depth: Awareness Only)

- ☐ Basic FEA awareness
- ☐ Load simulation basics
- ☐ Motion simulation basics

Project Checkpoints

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Design a gearbox housing
- ☐ Design a robot chassis
- ☐ Design a modular actuator mount

- ☐ Create a parametric enclosure system
- ☐ Build a full mechanical assembly in CAD

PHASE 5 — FABRICATION & PROTOTYPING

Workshop Skills

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Measuring tools
 - ☐ Vernier calipers
 - ☐ Micrometers
 - ☐ Steel rulers
 - ☐ Dial indicators
- ☐ Drilling
 - ☐ Drill press usage
 - ☐ Hole sizing
 - ☐ Countersinking
 - ☐ Tapping threads
- ☐ Cutting
 - ☐ Hacksaws
 - ☐ Angle grinders
 - ☐ Bandsaws
 - ☐ Rotary tools
- ☐ Finishing
 - ☐ Deburring
 - ☐ Sanding
 - ☐ Surface finishing

Rapid Prototyping

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ 3D printing
 - ☐ Printer setup
 - ☐ Slicer settings
 - ☐ Print orientation

- ☐ Support structures
- ☐ Laser cutting awareness
- ☐ CNC basics
- ☐ Foamboard prototyping
- ☐ Acrylic prototyping
- ☐ Quick mockup workflows

Manufacturing Awareness

(priority: High)

(depth: Working Knowledge)

- ☐ CNC machining basics
- ☐ Injection molding awareness
- ☐ Sheet metal basics
- ☐ Extrusion systems
- ☐ Waterjet awareness

Practical Tools

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ PrusaSlicer
- ☐ Cura
- ☐ Power drill
- ☐ Rotary tool
- ☐ Soldering tools
- ☐ Bench vise
- ☐ Clamps
- ☐ Hex key systems

Project Checkpoints

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Build a precision mechanical assembly
- ☐ Fabricate a robot frame
- ☐ Design and print custom brackets
- ☐ Manufacture a belt-drive system
- ☐ Build a compact actuator system

PHASE 6 — ACTUATORS & MOTION SYSTEMS

Motors & Actuators

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ DC motors
 - ☐ Torque-speed behavior
 - ☐ Gearmotors
 - ☐ Motor sizing
- ☐ Stepper motors
 - ☐ Step resolution
 - ☐ Torque characteristics
 - ☐ Motion precision
- ☐ Servo motors
 - ☐ Closed-loop motion
 - ☐ RC servos
 - ☐ Industrial servo awareness
- ☐ Linear actuators
 - ☐ Electric actuators
 - ☐ Stroke systems
 - ☐ Force ratings

Mechanical Motion Design

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Linkages
 - ☐ Four-bar mechanisms
 - ☐ Motion conversion
 - ☐ Mechanical amplification
- ☐ Cam systems
- ☐ Slider-crank systems
- ☐ Differential systems awareness
- ☐ Suspension basics

Robotics Mechanisms

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Wheeled robot mechanics
- ☐ Tracked systems
- ☐ Robot arms
- ☐ Grippers
- ☐ End effectors
- ☐ Pan-tilt systems

Precision Motion Systems

(priority: High)

(depth: Working Knowledge)

- ☐ Backlash reduction
- ☐ Repeatability
- ☐ Mechanical calibration
- ☐ Vibration minimization

Project Checkpoints

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Build a robot arm
- ☐ Build a belt-driven CNC axis
- ☐ Build a camera slider
- ☐ Build a precision gripper
- ☐ Build a pan-tilt tracking platform

PHASE 7 — MECHATRONIC STRUCTURE DESIGN

Chassis & Structural Systems

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Structural frames

- ☐ Modular structures
- ☐ Aluminum extrusion systems
- ☐ Compact packaging
- ☐ Load distribution
 - ☐ Center of gravity
 - ☐ Weight balancing
 - ☐ Structural reinforcement
- ☐ Shock & vibration management
 - ☐ Dampening systems
 - ☐ Isolation mounts
 - ☐ Resonance awareness

Enclosure Design

(priority: High)

(depth: Working Knowledge)

- ☐ Electronics integration
- ☐ Cable routing
- ☐ Ventilation
- ☐ Heat management
- ☐ Serviceability
- ☐ Waterproofing awareness

Thermal Mechanical Awareness

(priority: Medium)

(depth: Working Knowledge)

- ☐ Heat sinks
- ☐ Airflow basics
- ☐ Thermal expansion
- ☐ Passive cooling

Modular Engineering

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Swappable modules
- ☐ Maintenance-friendly design
- ☐ Rapid repair systems

- ☐ Upgradeable architecture

PHASE 8 — INTEGRATION WITH ELECTRONICS & CONTROL

Mechanical-Electrical Integration

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Motor mounting
- ☐ Sensor mounting
- ☐ Encoder integration
- ☐ Wire-safe mechanical design
- ☐ Cable chain systems

Sensor Mechanics

(priority: High)

(depth: Working Knowledge)

- ☐ Limit switches
- ☐ Hall sensors
- ☐ Optical sensor placement
- ☐ Mechanical alignment systems

Control-System Mechanics

(priority: High)

(depth: Working Knowledge)

- ☐ PID tuning intuition
- ☐ Mechanical response understanding
- ☐ Oscillation awareness
- ☐ System inertia awareness

Embedded Integration Awareness

(priority: High)

(depth: Working Knowledge)

- ☐ Mounting PCBs

- ☐ EMI-aware routing awareness
- ☐ Cooling electronics
- ☐ Battery mounting systems

PHASE 9 — DEBUGGING & FAILURE ANALYSIS

Mechanical Troubleshooting

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Misalignment diagnosis
- ☐ Friction diagnosis
- ☐ Vibration diagnosis
- ☐ Noise analysis
- ☐ Structural weakness identification

Precision Debugging

(priority: High)

(depth: Working Knowledge)

- ☐ Runout checking
- ☐ Tolerance mismatch diagnosis
- ☐ Backlash measurement
- ☐ Mechanical calibration

Reliability Engineering

(priority: High)

(depth: Working Knowledge)

- ☐ Wear reduction
- ☐ Lubrication basics
- ☐ Preventive maintenance
- ☐ Mechanical lifespan estimation

Failure Analysis

(priority: High)

(depth: Working Knowledge)

- ☐ Fatigue failures
- ☐ Thermal failures
- ☐ Fastener failures
- ☐ Bearing failures
- ☐ Structural cracking

PHASE 10 — SYSTEMS ENGINEERING & PRODUCT THINKING

Product-Oriented Mechanical Design

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Cost-aware design
- ☐ Manufacturability
- ☐ Assembly-line thinking
- ☐ Maintainability
- ☐ User-safe mechanisms

Startup-Oriented Mechanical Skills

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Rapid iteration workflows
- ☐ Prototype-to-production transition
- ☐ BOM optimization
- ☐ Vendor sourcing awareness
- ☐ Standard-part usage

Documentation & Workflow

(priority: High)

(depth: Working Knowledge)

- ☐ Mechanical BOM creation
- ☐ CAD organization
- ☐ Revision management
- ☐ Assembly documentation

Team Integration

(priority: High)

(depth: Working Knowledge)

- ☐ Working with electronics engineers
- ☐ Working with firmware developers
- ☐ Working with manufacturers
- ☐ Mechanical requirement communication

PHASE 11 — ADVANCED HIGH-ROI TOPICS

Robotics Mechanical Systems

(priority: High)

(depth: Working Knowledge)

- ☐ Differential drive systems
- ☐ Omniwheel systems
- ☐ Legged robot awareness
- ☐ Drone frame awareness

Automation Mechanics

(priority: High)

(depth: Working Knowledge)

- ☐ Conveyor systems
- ☐ Pick-and-place mechanics
- ☐ Linear automation systems
- ☐ Industrial fixture basics

Precision Engineering Awareness

(priority: Medium)

(depth: Awareness Only)

- ☐ Harmonic drives
- ☐ Ball screws
- ☐ Industrial linear rails
- ☐ Precision machining awareness

Pneumatics Awareness

(priority: Medium)

(depth: Working Knowledge)

- ☐ Pneumatic cylinders
- ☐ Air valves
- ☐ Air routing
- ☐ Compressor basics

Hydraulics Awareness

(priority: Optional)

(depth: Awareness Only)

- ☐ Hydraulic actuators
- ☐ Hydraulic pumps
- ☐ High-force systems awareness

PHASE 12 — PRACTICAL BUILD PORTFOLIO

Core Build Projects

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Build a 2-wheel robot chassis
- ☐ Build a robotic gripper
- ☐ Build a conveyor prototype
- ☐ Build a belt-drive linear rail
- ☐ Build a camera gimbal structure
- ☐ Build a CNC-style axis
- ☐ Build a compact gearbox
- ☐ Build a modular electronics enclosure

Integrated Mechatronics Projects

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Build a line-following robot

- ☐ Build a robotic arm
- ☐ Build an automated sorting mechanism
- ☐ Build a pan-tilt tracking system
- ☐ Build a mobile robot platform
- ☐ Build a sensor-actuated mechanism

Professional Capability Benchmarks

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Design manufacturable assemblies
- ☐ Create production-ready CAD
- ☐ Diagnose mechanical failures quickly
- ☐ Integrate mechanics with electronics
- ☐ Prototype systems rapidly
- ☐ Optimize mechanisms for cost/performance
- ☐ Build reliable repeatable systems

OPTIONAL / LOW-ROI TOPICS

Low Practical ROI Academic Topics

(priority: Optional)

(depth: Awareness Only)

- ☐ Advanced continuum mechanics
- ☐ Tensor-heavy solid mechanics
- ☐ Advanced thermodynamics derivations
- ☐ Complex analytical stress proofs
- ☐ Advanced fluid mechanics theory
- ☐ Advanced metallurgy
- ☐ Research-level finite element analysis
- ☐ Academic machine design proofs
- ☐ Advanced vibration mathematics
- ☐ Turbomachinery theory
- ☐ Niche manufacturing theory
- ☐ Research-level tribology

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PHASE 1 — CORE ELECTRICAL INTUITION

Electrical Fundamentals

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Voltage intuition
 - ☐ Potential difference
 - ☐ Voltage sources

- ☐ Supply rails
- ☐ Ground reference
- ☐ Current intuition
 - ☐ Electron flow intuition
 - ☐ Current paths
 - ☐ Current consumption
 - ☐ Short-circuit behavior
- ☐ Resistance intuition
 - ☐ Current limiting
 - ☐ Voltage drops
 - ☐ Heat generation
 - ☐ Pull-up/pull-down usage
- ☐ Power fundamentals
 - ☐ Watt calculations
 - ☐ Power dissipation
 - ☐ Efficiency intuition
 - ☐ Battery runtime estimation
- ☐ AC vs DC understanding
 - ☐ DC systems
 - ☐ AC mains basics
 - ☐ Frequency intuition
 - ☐ Safe AC handling awareness

Core Circuit Laws

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Ohm's Law
- ☐ Power equations
- ☐ Series circuits
- ☐ Parallel circuits
- ☐ Voltage division
- ☐ Current division
- ☐ Kirchhoff intuition
- ☐ Basic load calculations

Electrical Safety

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Low-voltage safety
- ☐ High-current hazards
- ☐ AC mains awareness
- ☐ Short-circuit prevention
- ☐ Fuse protection
- ☐ Reverse-polarity protection
- ☐ Battery safety
- ☐ Capacitor discharge awareness
- ☐ ESD awareness

Practical Mathematics

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Voltage calculations
- ☐ Current calculations
- ☐ Power calculations
- ☐ Resistance calculations
- ☐ Battery capacity calculations
- ☐ Efficiency calculations
- ☐ PWM duty-cycle calculations
- ☐ Sensor scaling calculations

PHASE 2 — ELECTRONIC COMPONENTS & HARDWARE

Passive Components

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Resistors
 - ☐ Resistance values
 - ☐ Power ratings
 - ☐ Tolerance
 - ☐ Pull-up/pull-down usage

- ☐ Capacitors
 - ☐ Ceramic capacitors
 - ☐ Electrolytic capacitors
 - ☐ Decoupling
 - ☐ Filtering basics
- ☐ Inductors awareness
 - ☐ Coils
 - ☐ Energy storage intuition
 - ☐ Noise filtering basics

Active Components

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Diodes
 - ☐ Rectifier diodes
 - ☐ Schottky diodes
 - ☐ Flyback diodes
 - ☐ LEDs
- ☐ Transistors
 - ☐ BJT basics
 - ☐ MOSFET basics
 - ☐ Switching circuits
 - ☐ Logic-level MOSFETs
- ☐ Voltage regulators
 - ☐ Linear regulators
 - ☐ Buck converters
 - ☐ Boost converters
 - ☐ Voltage stability
- ☐ Relays
 - ☐ Relay driving
 - ☐ Isolation basics
 - ☐ Inductive load handling

Sensors for Mechatronics

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Basic sensors
 - ☐ Temperature sensors
 - ☐ Light sensors
 - ☐ Pressure sensors
 - ☐ Distance sensors
- ☐ Motion sensors
 - ☐ Encoders
 - ☐ Hall sensors
 - ☐ Accelerometers
 - ☐ Gyroscopes
 - ☐ IMUs
- ☐ Robotics sensors
 - ☐ Ultrasonic sensors
 - ☐ IR sensors
 - ☐ LiDAR awareness
 - ☐ Camera module awareness
- ☐ Industrial sensors
 - ☐ Limit switches
 - ☐ Proximity sensors
 - ☐ Current sensors

Actuation Electronics

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ DC motor drivers
- ☐ H-bridges
- ☐ Stepper drivers
- ☐ Servo control systems
- ☐ Solenoid driving
- ☐ PWM motor control

PHASE 3 — PRACTICAL CIRCUIT BUILDING

Breadboarding & Prototyping

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Breadboard usage
- ☐ Power rail organization
- ☐ Jumper wire management
- ☐ Rapid prototyping workflows
- ☐ Modular testing

Soldering Skills

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Through-hole soldering
- ☐ SMD soldering basics
- ☐ Desoldering
- ☐ Wire joining
- ☐ Connector soldering
- ☐ Heat management

Wiring & Connectors

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Wire gauge basics
- ☐ Current capacity awareness
- ☐ Connector systems
 - ☐ JST
 - ☐ Dupont
 - ☐ XT60
 - ☐ Screw terminals
- ☐ Crimping
- ☐ Cable management
- ☐ Strain relief
- ☐ Shielded cable awareness

Practical Tools

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Multimeter

- ☐ Voltage measurement
- ☐ Current measurement
- ☐ Continuity testing
- ☐ Resistance testing
- ☐ Oscilloscope basics
- ☐ Bench power supplies
- ☐ Logic analyzer awareness
- ☐ Soldering station
- ☐ Hot air station awareness

Project Checkpoints

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Build LED driver circuits
- ☐ Build sensor-reading circuits
- ☐ Build a MOSFET switching system
- ☐ Build a motor-control prototype
- ☐ Build a regulated power distribution board

PHASE 4 — POWER SYSTEMS & BATTERIES

Battery Systems

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Lithium-ion batteries
- ☐ LiPo batteries
- ☐ Battery capacity
- ☐ C-rating understanding
- ☐ Charging systems
- ☐ Battery protection systems
- ☐ Battery balancing awareness

Power Distribution

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Voltage rails
- ☐ Current budgeting
- ☐ Grounding basics
- ☐ Common-ground systems
- ☐ Power isolation awareness

Voltage Conversion

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Buck converters
- ☐ Boost converters
- ☐ Buck-boost awareness
- ☐ Efficiency considerations
- ☐ Heat dissipation

Motor Power Systems

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Stall current awareness
- ☐ Motor startup surges
- ☐ Separate logic/motor power
- ☐ Noise suppression
- ☐ Flyback protection

Charging & Energy Systems

(priority: High)

(depth: Working Knowledge)

- ☐ USB power systems
- ☐ Battery charging ICs awareness
- ☐ Solar power awareness
- ☐ UPS basics

Project Checkpoints

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Build a battery-powered robot

- ☐ Build a power-distribution module
- ☐ Design safe charging workflows
- ☐ Build a multi-voltage power system

PHASE 5 — DIGITAL ELECTRONICS & EMBEDDED INTERFACING

Digital Logic Fundamentals

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Binary systems
- ☐ Logic levels
- ☐ HIGH vs LOW logic
- ☐ Boolean logic
- ☐ Pull-up/pull-down systems

Microcontroller Interfacing

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ GPIO usage
- ☐ Analog inputs
- ☐ PWM outputs
- ☐ Interrupt basics
- ☐ Timer awareness

Communication Protocols

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ UART
- ☐ I2C
- ☐ SPI
- ☐ CAN bus awareness
- ☐ USB awareness
- ☐ RS485 awareness

Embedded Hardware Platforms

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Arduino Uno
- ☐ ESP32
- ☐ Raspberry Pi 5 awareness
- ☐ STM32 awareness
- ☐ RP2040 awareness

Signal Conditioning

(priority: High)

(depth: Working Knowledge)

- ☐ Analog filtering
- ☐ Debouncing
- ☐ Noise reduction
- ☐ Voltage scaling
- ☐ Level shifting

Project Checkpoints

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Interface multiple sensors
- ☐ Control motors using PWM
- ☐ Build wireless sensor nodes
- ☐ Build a robotic controller board
- ☐ Integrate multiple communication protocols

PHASE 6 — PCB DESIGN & HARDWARE DEVELOPMENT

Schematic Design

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Reading schematics

- ☐ Drawing schematics
- ☐ Signal flow understanding
- ☐ Modular design

PCB Layout Fundamentals

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Component placement
- ☐ Trace routing
- ☐ Ground planes
- ☐ Decoupling capacitor placement
- ☐ High-current trace awareness

Practical PCB Design

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Connector placement
- ☐ Test points
- ☐ Mounting holes
- ☐ Modular interfaces
- ☐ EMI reduction basics

PCB Software

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ KiCad
- ☐ EasyEDA
- ☐ Altium Designer awareness

Manufacturing Workflow

(priority: High)

(depth: Working Knowledge)

- ☐ Gerber generation
- ☐ PCB ordering workflows
- ☐ BOM generation
- ☐ Assembly workflows

- ☐ PCB inspection

Project Checkpoints

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Design a sensor PCB
- ☐ Design a motor-driver PCB
- ☐ Design a robot-controller PCB
- ☐ Manufacture and assemble custom PCBs

PHASE 7 — ROBOTICS & MECHATRONICS ELECTRONICS

Robotics Power Architecture

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Power budgeting
- ☐ Motor/sensor isolation
- ☐ Battery distribution
- ☐ Noise-aware layouts

Motion-Control Electronics

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Encoder feedback
- ☐ Closed-loop motor awareness
- ☐ PWM tuning
- ☐ Stepper control systems

Sensor Fusion Awareness

(priority: High)

(depth: Working Knowledge)

- ☐ IMU integration
- ☐ Multi-sensor synchronization
- ☐ Data filtering basics

Autonomous System Electronics

(priority: High)

(depth: Working Knowledge)

- ☐ Wireless communication
- ☐ GPS awareness
- ☐ Camera integration awareness
- ☐ Edge-AI hardware awareness

Industrial Automation Basics

(priority: Medium)

(depth: Working Knowledge)

- ☐ PLC awareness
- ☐ Industrial relays
- ☐ Industrial sensors
- ☐ Industrial power supplies

PHASE 8 — DEBUGGING & TROUBLESHOOTING

Electrical Debugging

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Short-circuit diagnosis
- ☐ Open-circuit diagnosis
- ☐ Voltage-drop diagnosis
- ☐ Grounding issues
- ☐ Noise diagnosis

Power Debugging

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Brownout detection
- ☐ Current overload diagnosis
- ☐ Thermal issues

- ☐ Battery instability

Signal Debugging

(priority: High)

(depth: Working Knowledge)

- ☐ Oscilloscope signal reading
- ☐ PWM debugging
- ☐ Communication debugging
- ☐ Noise filtering diagnosis

Reliability Engineering

(priority: High)

(depth: Working Knowledge)

- ☐ Connector reliability
- ☐ Thermal reliability
- ☐ Vibration resistance
- ☐ Moisture awareness

Failure Analysis

(priority: High)

(depth: Working Knowledge)

- ☐ Burnt component diagnosis
- ☐ Reverse polarity damage
- ☐ MOSFET failures
- ☐ Regulator failures
- ☐ Battery failures

PHASE 9 — SYSTEMS INTEGRATION & PRODUCT THINKING

Mechatronic Systems Integration

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Electronics-mechanics integration

- ☐ Wiring-safe mechanical design
- ☐ Cable routing systems
- ☐ Sensor placement optimization

Product-Oriented Electronics

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Manufacturable electronics
- ☐ Cost optimization
- ☐ Modularity
- ☐ Repairability
- ☐ User-safe design

Startup-Oriented Hardware Skills

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Rapid prototyping
- ☐ Off-the-shelf module integration
- ☐ Vendor sourcing awareness
- ☐ BOM optimization
- ☐ Prototype-to-production workflows

Documentation & Workflow

(priority: High)

(depth: Working Knowledge)

- ☐ Wiring diagrams
- ☐ Connector labeling
- ☐ Revision tracking
- ☐ BOM management

PHASE 10 — ADVANCED HIGH-ROI TOPICS

Wireless Systems

(priority: High)

(depth: Working Knowledge)

- ☐ Wi-Fi basics
- ☐ Bluetooth basics
- ☐ RF module awareness
- ☐ Antenna placement basics

Signal Processing Awareness

(priority: Medium)

(depth: Working Knowledge)

- ☐ Filtering basics
- ☐ ADC basics
- ☐ DAC basics
- ☐ Sampling intuition

High-Power Electronics Awareness

(priority: Medium)

(depth: Awareness Only)

- ☐ AC motor drives awareness
- ☐ Inverters awareness
- ☐ Three-phase basics
- ☐ Industrial motor control awareness

Edge AI Hardware Awareness

(priority: Medium)

(depth: Awareness Only)

- ☐ AI accelerators awareness
- ☐ Vision-processing boards
- ☐ Embedded AI modules

Industrial Communication Awareness

(priority: Medium)

(depth: Awareness Only)

- ☐ Modbus awareness
- ☐ EtherCAT awareness
- ☐ Industrial CAN awareness

PHASE 11 — PRACTICAL BUILD PORTFOLIO

Core Electronics Projects

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Build sensor breakout systems
- ☐ Build motor-control systems
- ☐ Build battery-management systems
- ☐ Build wireless control systems
- ☐ Build power-monitoring systems

Mechatronics Projects

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Build a line-following robot
- ☐ Build a robotic arm controller
- ☐ Build a balancing robot
- ☐ Build an autonomous rover
- ☐ Build a smart actuator module

PCB Projects

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Design custom sensor PCBs
- ☐ Design integrated robot-controller boards
- ☐ Design motor-driver boards
- ☐ Manufacture and debug custom PCBs

Professional Capability Benchmarks

(priority: Extremely High)

(depth: Deep Mastery)

- ☐ Diagnose electronic failures quickly
- ☐ Design robust power systems
- ☐ Integrate sensors and actuators reliably
- ☐ Design manufacturable PCBs
- ☐ Build reliable embedded hardware

- ☐ Rapidly prototype mechatronic systems

OPTIONAL / LOW-ROI TOPICS

Low Practical ROI Academic Topics

(priority: Optional)

(depth: Awareness Only)

- ☐ Advanced electromagnetic field theory
- ☐ Tensor-heavy electromagnetics
- ☐ Microwave engineering
- ☐ Antenna theory derivations
- ☐ Advanced semiconductor physics
- ☐ Research-level analog IC design
- ☐ Advanced communication theory
- ☐ Power-grid engineering theory
- ☐ Advanced control-theory proofs
- ☐ Research-level VLSI design
- ☐ RF mathematics-heavy derivations

